UNIVERSITY OF CALGARY

Data 611 – Predictive Analysis (Winter 2025)

Project Report

Predicting Vehicle Insurance Premiums with Machine Learning

Group 6

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by Alvin Hui, Binh Nguyen, & Judy Kurupakorn

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1. Introduction

Transportation is an essential part of daily life, whether as a car owner or a pedestrian. Vehicles provide mobility and convenience, but they also come with risks, making safety and insurance crucial considerations. Car insurance plays a key role in protecting owners from financial loss due to accidents, covering both the insured vehicle and third parties involved.

A major concern for both individuals and businesses is determining whether the insurance premium they pay is reasonable. For households, selecting an insurance policy involves balancing affordability and coverage. For businesses, especially those managing fleets, assessing whether premium rates are appropriately set is critical for cost management and risk assessment.

This project aims to analyze factors influencing car insurance premiums using machine learning, specifically regression models. The dataset includes details such as insurance type, insured value, vehicle specifications (type, production year, number of seats, manufacturer, and usage), and the insurance period. By identifying key predictors of premium amounts, the model can provide insights for insurance companies, policymakers, and consumers, ensuring fair pricing and informed decision-making.

1.1 Research Questions

- How can we predict car insurance premiums using machine learning models?
- What are the key factors influencing car insurance premiums?

1.2 Objectives

To identify the most significant factors affecting vehicle insurance premiums, develop predictive models using regression techniques, evaluate and compare their performance, and summarize the results and key findings.

1.3 Background

Vehicle insurance is a contract that provides financial protection for vehicle owners against losses related to their vehicles (Infoassurance, 2019). The cost of this protection, known as the insurance premium, is influenced by various factors:

- **Driver characteristics:** Gender, age, and driving record impact premium calculations (The Personal, 2024; Yu et al., 2021).
- **Policy details:** The insurance type, coverage period, and effective year affect pricing (Infoassurance, 2019; The Personal, 2024).
- **Vehicle attributes:** The insured value, production year, make, type, seating capacity, and engine specifications influence premiums (Infoassurance, 2019; The Personal, 2024).
- **Usage factors:** How the vehicle is primarily used, and its carrying capacity is considered in premium calculations (Infoassurance, 2019; The Personal, 2024).
- **Claim history:** Previous claims paid affect future premiums (Colella & Jones, 2023; The Personal, 2024).

Vehicle insurance premium prediction involves using these factors to estimate the cost of insurance. Modern approaches include:

- Statistical models like Multiple Linear Regression and Generalized Linear Models (Colella & Jones, 2023; Y. Yang, 2025).
- Machine learning techniques such as Random Forest, Gradient Boosting (XGBoost), and Neural Networks (Y. Yang, 2025; Yu et al., 2021).
- Predictive modelling uses historical and current data to forecast future premiums (Phillips & Shang, 2002).

These methods aim to accurately assess risk and determine appropriate premium rates for each unique policyholder and vehicle combination.

2. Dataset

The comprehensive Vehicle Insurance dataset includes key attributes related to personal demographics, insurance details, and vehicle specifications, all essential for analyzing their impact on premium amounts.

- The dataset is sourced from Mendeley Data: Vehicle Insurance Data, originally obtained from the Ethiopian Insurance Corporation, contributed by Edossa Terefe (https://data.mendeley.com/datasets/34nfrk36dt/1).
- It consists of two CSV files covering 2011–2014 and 2014–2018.
- This project focuses on 2014–2018 data, which contains 16 columns and 508,499 rows.

2.1 Source and Accessibility

It is licensed under the Creative Commons Attribution 4.0 International (CC BY 4.0) license (https://creativecommons.org/licenses/by/4.0/), allowing users to share, modify, and utilize the data with proper attribution.

2.2 Variables with Description

The dataset consists of **16 variables**, with **Premium Amount (premium)** as the **dependent variable**, which is quantitative. The independent variables will be carefully selected during the modelling stage to ensure optimal predictive performance.

Column Name	Type	Data Description
SEX	Categorical	Gender of the insured individual: 0 = legal entity, 1 = male, 2 = female
INSR_BEGIN	Date	Policy start date
INSR_END	Date	Policy end date
EFFECTIVE_YR	Numerical	Year policy takes effect

Column Name	Type	Data Description
INSR_TYPE	Categorical	Insurance type: 1201 = private, 1202 = commercial, 1204 = motor trade road risk
INSURED_VALUE	Numerical	Value of insured item in USD
PREMIUM	Numerical	Premium amount in USD
OBJECT_ID	Numerical	Unique identification
PROD_YEAR	Numerical	Production year
SEATS_NUM	Numerical	Number of seats 0–198
CARRYING_CAPACITY	Numerical	Item's weight/load limit 0–224000
TYPE_VEHICLE	Categorical	Category of insured vehicle i.e. Pick-up, Truck, Bus
CCM_TON	Numerical	Engine displacement or weight capacity
MAKE	Categorical	Manufacturer i.e. NISSAN, TOYOTA
USAGE	Categorical	Vehicle usage, i.e. Own Goods, General Cartage, Private, Own service, Taxi
CLAIM_PAID	Numerical	Previous claim amounts paid

2.3 Data Cleaning and Data Preparation

The initial raw data were retained in their entirety to ensure the completeness of the dataset, allowing for the proper selection of all available features during individual modelling stages.

The data preparation process included the following steps:

2.3.1 Handling missing values

- Check for duplicate rows: If duplicates were found, they were dropped to ensure data integrity.
- Check for NaN values: If the percentage of NaN values in a column was less than 5%, those records were dropped
- Check for zero values:
    - Those records were dropped if the percentage of zero values in a column was less than 5%.
    - After addressing duplicate and NaN rows, we found a total of 231,303 rows with zero values in the insured_value column. These zero values were replaced with the median value, which was calculated based on the vehicle's manufacturer, type of vehicle, and production year. The replacement condition required that the zero-value rows match the manufacturer and type of vehicle, along with the closest matching production year from the reference data frame created using non-zero insured_value records.

2.3.2 Encoding Categorical Variables

Categorical variables were revised to character values for better readability and analysis.

- insr_type: 1201: "private", 1202: "commercial", 1204: "motor trade road risk"
- sex: 0: "legal entity", 1: "male", 2: "female"

2.3.3 Feature Engineering

Adding new columns:

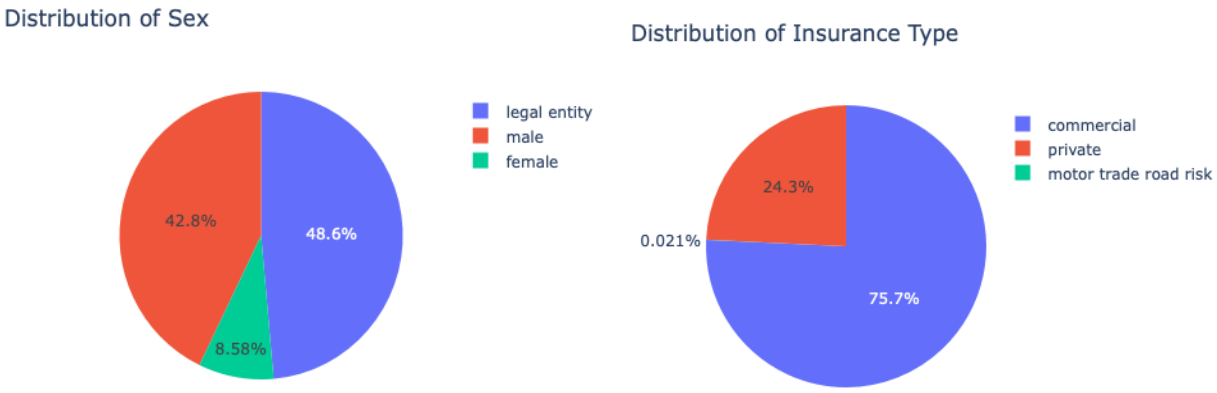
- insurance_duration: derived by calculating the difference between insr_end and insr_begin and converting it into days.
- insurance_claim: a binary column to indicate whether a claim was made.
    - If the claim_paid column had a value, the column was set to “claimed”.
    - If the claim_paid column was empty, the column was set to “no claim”.

2.4 The Exploratory Data Analysis (EDA)

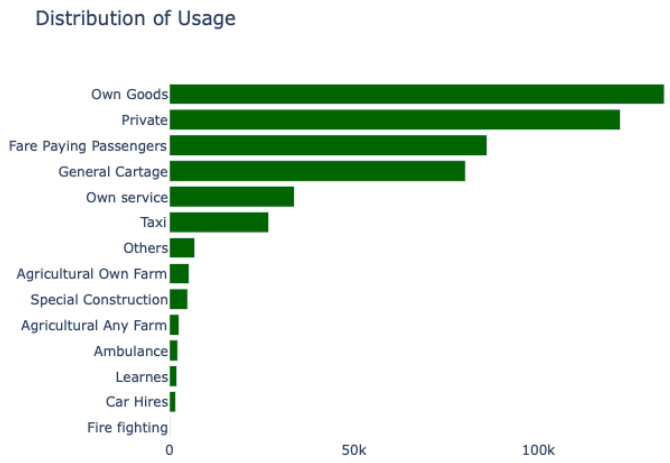
This section comprehensively examines the dataset to uncover underlying patterns, relationships, and potential factors influencing premium amounts. Through statistical and visual analysis. The exploration focuses on:

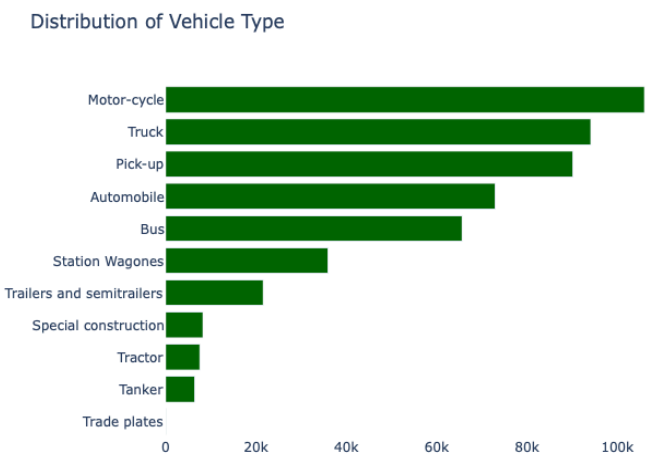
- Understanding the distribution and characteristics of vehicle insurance
- Examining relationships between premium amount and various factors

The following figures provide on selecting numerical and categorical features from the dataset. For the numerical features, statistical summaries are provided for key variables such as insured value, premium amount, production year, and claim paid amount. Additionally, the distribution of categorical features, including gender, insurance type, vehicle type, and car usage, is visualized using bar charts to highlight patterns and trends.



The first chart shows that legal entities make up the largest segment at 48.6%, followed by males at 42.8%, and females at 8.58% of the customer base. The second chart indicates that commercial insurance heavily dominates at 75.7% of policies, while private insurance accounts for 24.3%. A tiny fraction (0.021%) for motor trade road risk insurance exists.

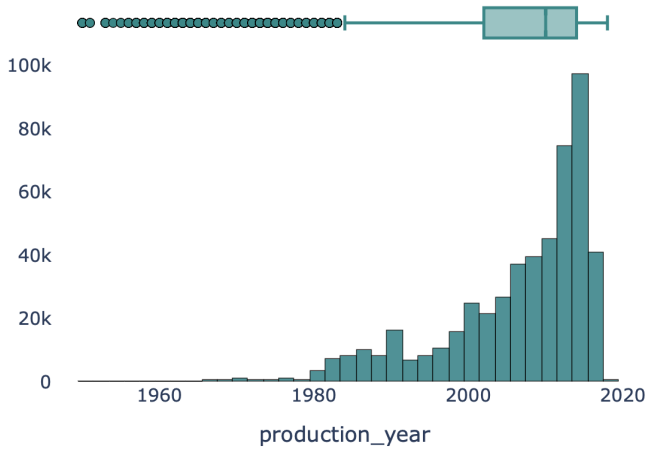




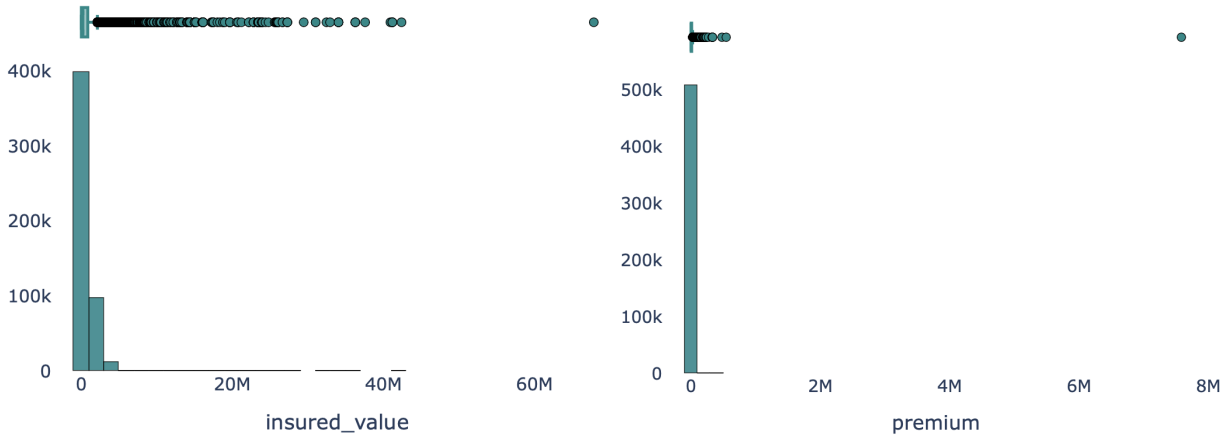
The first bar chart shows various vehicle usages, with "Own Goods" being the most common, followed by "Private" and "Fare Paying Passengers". Other categories like "Fire Fighting" and "Car Hires" have much lower usage. The second bar chart displays different types of vehicles, with "Motor-cycle" being the most prevalent, followed by "Truck" and "Pick-up". Less common types include "Trade Plates" and "Tanker".

	insured_value	premium	production_year	claim_paid
count	503,800	503,800	503,800	37,705
mean	779,240	8,007	2,007	243,412
std	989,392	15,567	9	1,082,849
min	1	4	1,958	0
25%	150,000	667	2,002	14,900
50%	480,000	3,539	2,010	36,824
75%	1,100,000	10,679	2,014	135,523
max	67,824,388	7,581,230	2,018	79,247,141

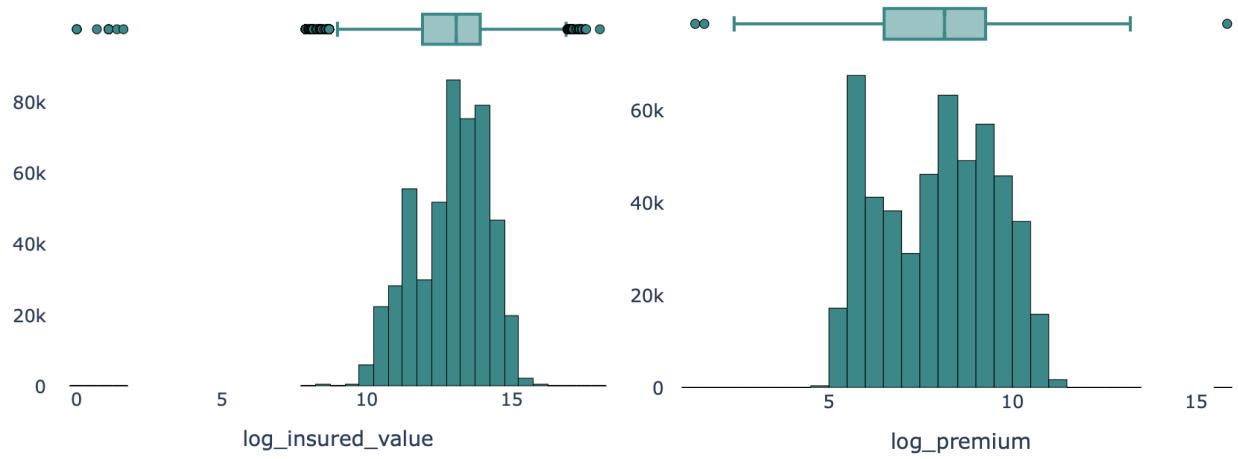
statistical summaries: insured_value, premium, production_year, claim_paid



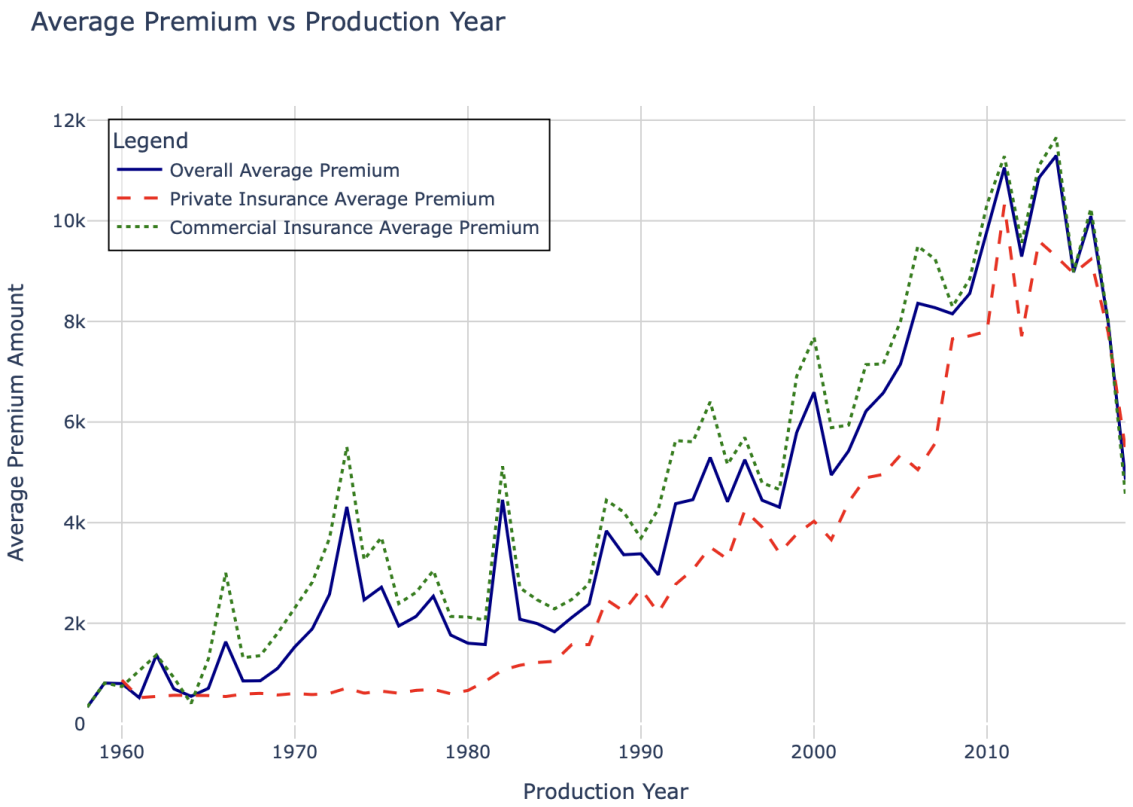
Distribution of production year indicates the frequency of production years, with most vehicles produced between 2000 and 2020. The box plot highlights the spread and outliers of the production year data, showing the interquartile range (IQR) from around 2002 to 2014.



Distribution of Insured Value and Premium Amount: A majority of data points are clustered around lower values, with a few outliers extending to much higher values. Further action may be required to reveal the data pattern.



The initial histogram of insured value and premium amount did not reveal clear data patterns. To enhance interpretability, a log transformation was applied, which spread out smaller values and compressed larger ones. The box plot on the left shows a wide range of insured values, with outliers on the lower end. The corresponding histogram indicates that most data points are concentrated at higher values, with a noticeable spike at the lower end. For the log-transformed premium amount, the box plot exhibits a more symmetric distribution with fewer extreme outliers. The histogram of log premiums appears more normally distributed, with the majority of data points centred around the middle range of log-transformed values.



The analysis shows a general upward trend in average insurance premiums over time. Vehicles produced after 2000 experienced a sharp increase in premiums, peaking in recent years before slightly declining. In contrast, older vehicles (manufactured before 1980) had more stable and lower premium amounts. Additionally, commercial insurance premiums tend to be higher than private insurance premiums. This is likely due to several factors, including increased usage, higher mileage, and greater exposure to risk. They may also transport valuable goods or passengers, further increasing liability and potential claim costs.

The data suggests that legal entities and commercial insurance dominate the market, with vehicle type, usage, and production year influencing premiums. Insured value and premium amounts may have non-linear relationships, with higher values leading to sharp premium increases. Log transformation improves data distribution, revealing clearer patterns. For regression modelling to capture complex relationships and optimize predictions, key steps include feature selection, handling outliers, and applying regularization techniques.

3. Analysis

In this study, Random Forest and Multiple Linear Regression (MLR) have been selected to model vehicle insurance premiums, aiming to identify key influencing factors and develop an accurate predictive model. MLR is chosen for its interpretability and ability to quantify relationships between predictors and premiums, assuming linearity, independence, and normally distributed residuals. Random Forest, a powerful ensemble method, is selected for its ability to capture complex, nonlinear relationships and its robustness against overfitting.

To enhance model performance, we fine-tune hyperparameters, applying regularization in MLR and optimizing tree depth and feature selection in Random Forest. Model performance is evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2). Finally, the comparison evaluates both models to determine the most effective approach for predicting vehicle insurance premiums.

3.1 Random Forest

3.1.1 Introduction to Random Forest

Random Forest is a supervised machine learning algorithm that leverages ensemble learning, a technique that combines predictions from multiple decision trees and delivers robust results for both classification and regression tasks. In this project, we use a Random Forest Regressor for two goals:

- Predict car insurance premiums
- Identify the key factors influencing car insurance premiums

Why Random Forest is Suitable for Our Project

In this project, we are building a model to predict the insurance premiums. It is a typical regression problem, and Random Forest is particularly suitable for this type of regression task due to its ability to aggregate predictions from multiple decision trees, handle both categorical

variables (e.g., TYPE_VEHICLE) and numerical variables (e.g., INSURED_VALUE), and reduce error variance.
Additionally, the Random Forest model quantifies the contribution of each predictor through feature importance scores, directly addressing the objective of our project and making it the ideal model for our analysis.

3.1.2 Flow of Model Development

The model development flow follows a structured modelling-building workflow from data cleaning to modelling, evaluation, hyperparameter tuning, and final model building.

The dataset first undergoes data preprocessing tasks, such as handling missing values and encoding categorical variables, etc. The cleaned data is then partitioned into training and testing sets (80-20 split).

A baseline Random Forest Regressor is then initialized with default hyperparameters to establish a performance benchmark.

Next, potential issues such as overfitting or underfitting are identified. To address these, hyperparameter tuning is conducted via randomized search with cross-validation, optimizing parameters like tree depth (max_depth) and the number of trees (n_estimators), etc. The optimized model is re-evaluated to quantify improvements in prediction accuracy and generalization.

Finally, feature importance scores from the tuned model are analyzed to identify the most influential predictors.

3.1.3 Basic Random Forest Model

In this step, a baseline Random Forest model was trained using default hyperparameters to establish performance benchmarks.

The dataset was first partitioned into an 80-20 split for training and testing, ensuring reproducibility with a fixed random state. Predictors included insured value, vehicle type, usage, production year, seats, engine capacity (CCM), and insurance type, which were selected for their logical relevance to insurance premium calculations. Categorical variables (e.g., vehicle_type, usage) were one-hot encoded, while numerical features (e.g., production year) were retained in their raw form.

Baseline Model Performance

A baseline Random Forest Regressor was trained using default hyperparameters. Performance metrics show a significant gap between training and testing errors, indicating overfitting:

Metric	**Training Set**	**Testing Set**	**Difference (%)**
RMSE	4,009.18	6,976.30	**+73.9%**
MAE	2,084.79	4,045.05	**+94.0%**
R^2	90.6%	71.6%	**-21.0%**

Interpretation of Results

1. Overfitting indicated by RMSE, and MAE

The large difference between training and testing errors (e.g., 73.9% difference in RMSE) suggests substantial overfitting. The model performs very well on training data but fails to generalize to unseen data.

2. Overfitting indicated by R^2 Scores

The model explains approximately 91% of the variance in training premiums, but drops to around 72% for testing data, indicating overfitting.

The substantial overfitting (as shown by the large gap between training and test results) highlights the need for hyperparameter tuning to improve generalization.

3.1.4 Hyperparameter Tuning

Objective and Approach

For our Random Forest model, hyperparameter tuning optimization is used to mitigate overfitting observed in the baseline model (e.g., (e.g., 73.9% gap in RMSE between training and testing sets). We conducted a randomized search over a predefined hyperparameter space, evaluating 30 combinations with 3-fold cross-validation to ensure robust performance estimation.

Hyperparameters Tested

Different combinations of hyperparameters were tested in order to increase the accuracy and better generalize the model.

Key hyperparameters tested in our Random Forest Model include:

Hyperparameter	**Values Tested**	**Purpose**
n_estimators	[200, 300, 400]	Controls the number of trees. Higher values reduce variance but increase computation.
max_depth	[None, 15, 20]	Limits tree depth. Smaller values prevent overfitting.
min_samples_split	[10, 20]	Minimum samples required to split a node. Higher values simplify trees.
min_samples_leaf	[3, 5, 10]	Minimum samples required at a leaf node. High values reduce overfitting.
max_features	["sqrt", 0.3, 0.5]	Features considered per split. Lower values decorrelate trees.
max_samples	[0.6, 0.8, None]	Bootstrap sample size.

Throughout the model building, randomized search was chosen over grid search for computational efficiency, as it allows us to explore a broad hyperparameter combination without exhaustive iteration.

Optimized Model Performance

After tuning, the model achieved the following performance with the best hyperparameters

Metric	Training Set	Testing Set	Difference (%)
RMSE	6,109.13	6,641.06	**+8.7%**
MAE	3,483.67	3,864.53	**+10.9%**
R^2	78.3%	74.2%	**-5.2%**

Interpretation of Results:

1. Reduced Overfitting

The hyperparameter tuning process significantly narrowed the gap between training and testing errors, leading to better generalization. For example, the RMSE difference decreased from 73.9% (baseline model) to 8.7% (tuned model). The gaps between MAE and R^2 have also been narrowed. This improvement suggests that the model no longer memorizes noise in the training data and instead focuses on learning meaningful patterns, making it more reliable for predicting unseen insurance policies.

2. Balanced Bias-Variance Tradeoff

The tuning process also led to a more balanced tradeoff between bias and variance. The training R^2 decreased from 90.6% to 78.3%, indicating a slight increase in bias that prevents overfitting. Meanwhile, the testing R^2 improved from 71.6% to 74.2%, demonstrating reduced variance and better generalization. This shift shows that the model now prioritizes capturing generalizable patterns rather than fitting outliers, resulting in a more stable predictive model.

Results Comparison and Improvements

Model performance before and after hyperparameter tuning:

Before Hyperparameter Tuning

Metric	**Training Set**	**Testing Set**	**Difference (%)**
RMSE	4,009.18	6,976.30	**+73.9%**
MAE	2,084.79	4,045.05	**+94.0%**
R^2	90.6%	71.6%	**-21.0%**

After Hyperparameter Tuning

Metric	**Training Set**	**Testing Set**	**Difference (%)**
RMSE	6,109.13	6,641.06	**+8.7%**
MAE	3,483.67	3,864.53	**+10.9%**
R^2	78.3%	74.2%	**-5.2%**

The hyperparameter tuning process successfully reduced overfitting, as shown in the significant decrease in the training-testing RMSE, MAE, and also R^2 gap. In the baseline model, the RMSE difference between training and testing was 73.9%, indicating that the model fit the training data too closely while failing to generalize. After tuning, this gap was reduced to 8.7%, demonstrating a better generalization of unseen data.

Additionally, the results reflect a more balanced tradeoff between bias and variance. While the training R^2 dropped from 90.6 to 78.3, indicating a slight decline in model complexity, the testing R^2 improved from 71.6 to 74.2, confirming that the model generalizes better. These changes suggest that the tuned model no longer overfits the training data but instead achieves a balance, making it more reliable for predictions on new data.

3.1.5 Feature Importance

The second research goal of our project is to determine the factors most influential to insurance premiums. This information can be obtained from the feature importance scores. This score quantifies the contribution of each variable to the Random Forest model's predictions, measured by the mean reduction in impurity across all decision trees.

By examining how much each predictor contributes to the model's decision-making, we gain insights into the most influential variables in premium estimation.

The top five most important features identified by the tuned Random Forest model are:

Rank	**Feature**	**Importance Score**
1	Insured Value	0.6582
2	Engine Capacity (Ton)	0.1115
3	Production Year	0.0569
4	Vehicle Usage	0.0428
5	Number of Seats	0.0317

Interpretation of Results:

1. Insured Value is the dominant factor (65.8%):

With an importance score of 0.6582, indicating that premium amounts are highly correlated with the vehicle's insured value. This aligns with industry expectations, as higher insured values lead to higher premiums.

2. Engine Capacity (Ton) (11.1%):

Engine capacity contributes 0.1115 to the model's predictions. Larger engines typically imply higher replacement costs and greater risk, leading to increased premiums.

3. Production Year (5.7%):

With a score of 0.0569. Newer vehicles tend to have higher insured values, but they may also come with advanced safety features, potentially lowering risk.

4. Vehicle Usage (4.3%):
Commercial vehicles, especially those used for fare-paying passengers, often have higher accident risks, impacting premium pricing.

5. Number of Seats (3.2%):
Larger vehicles (e.g., buses, vans) may have different risks compared to smaller passenger cars.
The feature importance results align well with industry expectations. Risk exposure plays a crucial role in determining the premium amount. Higher insured values indicate more expensive vehicles, which lead to higher claims in case of accidents. Similarly, vehicles used for commercial purposes or fare-paying passengers generally face greater risk exposure, resulting in higher premiums.

3.1.6 Summary of Analysis

In this project, we developed a Random Forest Regressor to predict car insurance premiums and identify key pricing factors. The model was trained on vehicle attributes like insured value, engine capacity, production year, usage, and number of seats.

The baseline model showed overfitting, with a 73.9% RMSE gap between training and testing. To improve generalization, we then conducted hyperparameter tuning using Randomized Search with Cross-Validation, reducing the RMSE gap to 8.7% and improving the bias-variance tradeoff.

Feature importance analysis confirmed insured value (65.8%) as the top predictor, followed by engine capacity (11.1%), production year (5.7%), vehicle usage (4.3%), and number of seats (3.2%). These results align with industry experience that premiums are driven by risk exposure and vehicle specifications.

3.2 Multiple Linear Regression

3.2.1 Introduction to Multiple Linear Regression (MLR)

In this project, we apply MLR to achieve two primary goals:

- Predict vehicle insurance premiums based on policyholder attributes, vehicle characteristics, and claims history. **Dependent variable:** Vehicle insurance premium.
- Identify key factors influencing insurance premiums, providing insights for pricing models and risk assessment. **Independent variables:** Policyholder attributes (sex, insurance_type, log_insured_value), vehicle characteristics (production_year, vehicle_type), usage, and claims history (insurance_claim).

This project aims to predict vehicle insurance premiums, which is a typical regression problem. MLR is well-suited for this analysis due to the following reasons:

- **Interpretability:** MLR provides clear insights into the effect of each predictor on the insurance premium, which is crucial for making informed business decisions.
- **Handling of mixed data types:** MLR can accommodate both categorical predictors (e.g., insurance_type, vehicle_type) and numerical predictors (e.g., log_insured_value, production_year).
- **Efficiency in large datasets:** Unlike more complex machine learning models, MLR is computationally efficient and does not require extensive tuning.

3.2.2 Flow of Model Development

The model development process follows a structured workflow, progressing through data partitioning, model fitting, model evaluation, and model optimization.

The dataset first undergoes preprocessing, including handling missing values and encoding categorical variables. Once cleaned, the data is partitioned into training (80%) and testing (20%) sets to ensure proper model validation and prevent overfitting.

An additive Multiple Linear Regression (MLR) model is first established to serve as a benchmark. Ordinary Least Squares (OLS) regression is applied to estimate coefficients and evaluate predictor significance.

Next, the model undergoes assumption testing to validate key regression assumptions:

- **Linearity** is confirmed using the Ramsey RESET test.
- **Normality of Residuals** is checked using the Shapiro-Wilk test and skewness/kurtosis measures.
- **Homoscedasticity** is assessed via the Breusch-Pagan and White's tests.
- **Multicollinearity** is diagnosed with Variance Inflation Factor (VIF) analysis.
- **Independence of Errors** is skipped because the dataset is not time-series data.

Following assumption validation, model optimization is performed to enhance predictive accuracy:

- **Feature Selection** is conducted using stepwise selection to retain the most relevant predictors.
- **Polynomial and Interaction Terms** are explored to capture potential nonlinear relationships.
- **Hyperparameter Tuning** is performed using Lasso, Ridge, and Elastic Net regularization techniques to balance bias-variance trade-offs and improve model generalization.

To quantify model performance, predictive evaluation metrics are calculated for both the training and test datasets:

- **R-squared (R^2)** is used to evaluate the fit of a model while taking into account the number of predictors included.
- **Root Mean Squared Error (RMSE)** to assess prediction accuracy in the original unit.
- **Mean Absolute Error (MAE)** to determine the average magnitude of prediction errors.

Finally, after model tuning and evaluation, the most influential predictors are analyzed to interpret their impact on insurance premium pricing. The insights gained from this analysis help insurers refine their pricing strategies while providing policyholders with guidance on factors influencing premium costs.

3.2.3 Model Fitting

This analysis refines a predictive model for vehicle insurance premiums through iterative regression modelling. We began with stepwise selection and an additive MLR model, identifying underfitting and multicollinearity. Assumption testing guided improvements, including outlier removal, robust standard errors, and interaction/polynomial terms, though overfitting emerged. Regularization (Lasso, Ridge, Elastic Net) addressed stability, with Ridge selected for hyperparameter tuning due to its balanced performance. Despite some MLR assumption violations, this approach improved generalization and predictive accuracy.

Additive Model

The additive MLR model showed mixed performance, with potential underfitting. RMSE dropped 41.97% from training (15,100.52) to testing (8,762.19), indicating better test performance but poor training fit. MAE remained stable (-1.70%), suggesting consistent prediction errors. The R^2 score improved significantly (+116.82%), with a higher test R^2 (39.83%) than training (18.37%), an unusual sign of underfitting. These results highlight multicollinearity and assumption violations, warranting further refinement through feature selection, assumption testing, and model adjustments like interaction terms or regularization.

Metric	Training Set	Testing Set	Difference (%)
RMSE	15,100.52	8,762.19	**-41.97%**
MAE	4,728.66	4,648.09	**-1.70%**
R^2	18.37%	39.83%	**+116.82%**

Additive model after dropping influential outliers and using robust standard errors

Removing influential outliers and applying robust standard errors improved model stability. RMSE dropped significantly on training (8,004.63) but slightly increased on testing (+7.25%), while MAE rose moderately (+5.16%). R^2 improved to 46.16% (train) but dropped slightly to 42.24% (test, -8.49%), indicating reduced underfitting but slight overfitting. The model generalizes better but still needs refinement, particularly in addressing multicollinearity and feature selection.

Metric	Training Set	Testing Set	Difference (%)
RMSE	8,004.63	8,585.18	**+7.25%**
MAE	4,330.44	4,554.07	**+5.16%**
R^2	46.16%	42.24%	**-8.49%**

Interaction model after dropping influential outliers and using robust standard errors

Introducing interaction terms significantly improved training performance (R^2 = 52.21%, RMSE = 7,541.89), but caused severe overfitting, with testing R^2 dropping to -55.98% and RMSE increasing by 87.06%. The MAE increase (+7.63%) indicates unstable predictions. These results suggest that multicollinearity and overfitting from interaction terms degrade generalization, requiring regularization or refined feature selection.

Metric	Training Set	Testing Set	Difference (%)
RMSE	7,541.89	14,107.61	**+87.06%**
MAE	4,024.34	4,331.47	**+7.63%**
R^2	52.21%	-55.98%	**-207.22%**

Polynomial model after dropping influential outliers and using robust standard errors (with insured value squared)

Adding a squared term for insured value did not significantly improve the model. While training R^2 remained at 46.15%, the testing R^2 dropped to 28.66% (-207.22%), indicating overfitting. RMSE increased by 87.06% on the test set, and MAE rose by 7.63%, showing reduced generalization. Regularization (Ridge/Lasso) is needed to control complexity and improve stability.

Metric	Training Set	Testing Set	Difference (%)
RMSE	8,005.67	9,540.70	**+19.17%**
MAE	4,246.61	4,512.21	**+6.25%**
R^2	46.15%	28.66%	**-37.90%**

3.2.4 Hyperparameter Tuning

Lasso, Ridge, and Elastic Net were applied for regularization to address overfitting and improve generalization. These methods help reduce coefficient instability, handle multicollinearity, and enhance model interpretability by penalizing large coefficients. Since adding interaction and polynomial terms can lead to overfitting, hyperparameter tuning was performed using X_{train} and y_{train} from the additive filtered model, which excluded influential outliers. Additionally, robust standard errors were used to ensure more reliable estimates.

Model performance comparison:

- Ridge performed best, achieving the highest train R^2 (46.16%) and test R^2 (42.24%), with lower RMSE (8004.64 train, 8585.17 test) and stable MAE.
- Lasso selected fewer features (20) but had a slightly lower test R^2 (42.11%), indicating possible underfitting.
- Elastic Net performed similarly to Lasso but kept more features (27), balancing Lasso's sparsity and Ridge's stability.

Lasso, Ridge, Elastic Net

Model	Best Alpha	Selected Features	Train R^2	Test R^2
Lasso	0.01	20	45.69%	42.11%
Ridge	**1.00**	**27**	**46.16%**	**42.24%**
Elastic Net	0.01	27	45.67%	41.92%

Model	Train RMSE	Test RMSE	Train MAE	Test MAE
Lasso	8039.77	8594.47	4321.79	4543.91
Ridge	**8004.64**	**8585.17**	**4330.44**	**4554.07**
Elastic Net	8041.52	8608.54	4338.84	4561.25

Final decision: Ridge regression

Ridge was selected due to its higher R^2, lower RMSE, and retention of more features, making it the best option for improving model performance while maintaining generalization.

3.2.5 Summary of Analysis

This section applied Multiple Linear Regression (MLR) to predict vehicle insurance premiums and identify key influencing factors. The initial additive model revealed issues such as underfitting and multicollinearity, necessitating further refinements. Removing influential outliers and applying robust standard errors improved model stability, but interaction and polynomial terms led to severe overfitting. To address these challenges, regularization techniques (Lasso, Ridge, and Elastic Net) were implemented, with Ridge regression emerging as the optimal choice due to its balanced performance, higher R^2, and lower RMSE while retaining more predictive features.

Despite some remaining assumption violations, the final Ridge model provided better generalization, demonstrating improved predictive accuracy and reduced overfitting. This iterative approach highlights the importance of assumption testing, feature selection, and regularization in refining regression models for insurance pricing and risk assessment.

3.3 Comparison of Predictive Performance: MLR vs. Random Forest Models

Both Random Forest and Multiple Linear Regression (MLR) were applied to predict vehicle insurance premiums and identify key influencing factors. While MLR provides interpretability and a clear understanding of predictor relationships, Random Forest offers greater predictive accuracy by capturing complex, nonlinear feature interactions.

3.3.1 Performance Comparison

After hyperparameter tuning, Random Forest outperformed MLR across all evaluation metrics:

Model	**Train R^2**	**Test R^2**	**Train RMSE**	**Test RMSE**	**Train MAE**	**Test MAE**
Random Forest (tuned)	78.30%	74.20%	6,109.13	6,641.06	3,483.67	3,864.53
MLR (Ridge)	46.15%	28.66%	8,005.67	9,540.70	4,246.61	4,512.21

- Higher R^2: Random Forest explains 74.2% of the variance in test data, compared to 42.24% for MLR.
- Lower RMSE: Random Forest has a 22.6% lower test RMSE than MLR, indicating more accurate predictions.
- Lower MAE: Random Forest also has a lower MAE, showing reduced average prediction error.

3.3.2 Key differences & advantages of Random Forest

- Better predictive power: Random Forest captures complex relationships, while MLR struggles with assumption violations (e.g., multicollinearity, nonlinearity).
- Reduced overfitting: Hyperparameter tuning narrowed the generalization gap for Random Forest (Test R^2 only 5.2% lower than Train R^2), whereas MLR showed a larger gap between training and testing performance.
- Feature importance insights: Random Forest identifies key factors influencing insurance premiums, with Insured Value (65.8%) as the most significant, followed by Engine Capacity, Production Year, and Vehicle Usage. MLR lacks the flexibility to handle complex relationships among features.

3.3.3 Conclusion

While MLR provides interpretability, Random Forest delivers superior predictive performance for vehicle insurance premium estimation. With higher R^2, lower RMSE, and reduced error variance, Random Forest is the more reliable model for this dataset, making it the preferred choice for accurate premium prediction and risk assessment.

4. Discussion of insights and findings

This section explores the key findings derived from the analysis, highlighting the most influential factors in insurance premium pricing and comparing the performance of different modelling approaches.

4.1 Key Findings

1. Most Impactful Variables on Insurance Premium

- Insured Value (65.8% importance) dominated premium calculations, aligning with industry norms where higher coverage correlates with higher premiums.
- Engine Capacity (11.1% importance) and Production Year (5.7% importance) were secondary drivers, reflecting repair costs and depreciation risks.
- Vehicle Usage (4.3% importance) highlighted elevated risks for commercial vehicles, such as taxis, delivery vans, etc.

2. Random Forest Outperforms MLR in Complexity Handling

- The Random Forest model demonstrated better performance in capturing nonlinear relationships between predictors and insurance premiums, whereas MLR's linear assumptions likely limited its accuracy.
- Overfitting was successfully mitigated in Random Forest through hyperparameter tuning, reducing the RMSE gap between training and testing from 73.9% to 8.7%, ensuring reliable generalization to the unseen insurance policy.

3. Additional Insights

- Bias-Variance Tradeoff is observed through model tuning, which shifts the model from memorizing training data (training R^2 = 90.6%) to generalizing patterns (testing R^2 = 74.2%).
- Feature importance analysis results align with industry practice where insurance premium tends to tight with risk exposure.

4.2 Challenges Encountered

4.2.1 Random Forest

The challenges encountered when building the random forest model are as follow:

a. Data Quality

Missing values and inconsistent data formats required extensive cleaning. Another challenge was determining the proper data imputation method for each feature (such as missing Premium values, Vehicle type, etc.). These challenges were addressed through a detailed study of each feature's characteristics and group discussions to identify the most appropriate imputation methods.

b. Computation Time

Hyperparameter tuning with more than 500k rows demanded significant computational resources, especially when cross-validation was involved. This issue was mitigated through randomized search, data partitioning, and parallel processing.

4.2.2 Multiple Linear Regression

During the Multiple Linear Regression (MLR) analysis, several challenges emerged, impacting model performance and interpretability. These challenges were primarily related to data quality issues, computational efficiency, and model assumptions.

a. Data quality issues

- **Multicollinearity:** High correlation among predictors (e.g., insured value and policy type) led to unstable coefficient estimates. Variance Inflation Factor (VIF) analysis flagged multicollinearity, requiring feature selection and regularization to mitigate its impact.
- **Outliers & Influential points:** The presence of extreme values distorted coefficient estimates, affecting predictive accuracy. Cook's Distance analysis helped identify influential points but removing them altered model stability, requiring robust standard errors.

- **Missing or skewed data:** Some categorical variables had imbalanced distributions, affecting coefficient estimation. While missing values were handled during preprocessing, skewed distributions may have contributed to assumption violations.

b. Model Assumption Violations & Interpretability

- **Linearity & Normality violations:** Residual analysis indicated that the relationship between predictors and premiums was not strictly linear, affecting model fit. Polynomial terms and interaction effects were introduced, but they led to overfitting.
- **Heteroscedasticity:** Unequal variance in residuals affected the reliability of coefficient significance. Robust standard errors were applied, but heteroscedasticity remained an issue, impacting confidence in the model's predictions.
- **Diminishing interpretability with complexity:** Introducing polynomial and interaction terms improved training performance but severely degraded test accuracy, making the model less interpretable and leading to overfitting.

c. Computational & Optimization Challenges

- **The high computational cost of regularization:** Applying Lasso, Ridge, and Elastic Net for feature selection and bias-variance tradeoff optimization was computationally expensive, particularly when tuning hyperparameters. Ridge regression was chosen for its balance between complexity and performance.
- **The trade-off between bias and variance:** While regularization helped reduce overfitting, it also introduced a trade-off by shrinking some coefficients too aggressively, potentially removing meaningful predictors.

4.3 Limitations

4.3.1 General Limitation

- **Assumptions and Biases**

  While Random Forest does not assume linearity, it may still exhibit bias if the dataset does not represent the broader policyholder population.

- **Regional Bias**

  The dataset is sourced from a single insurer (Ethiopian Insurance Corporation), meaning patterns may not generalize to other markets with different risk profiles (e.g., traffic laws, and vehicle types).

- **Limited Features**

  Currently, the dataset includes 16 features as predictors of the insurance premium. However, adding variables such as driver age, accident history, or geographic risk zones could improve model accuracy.

4.3.2 MLR

- **Assumption violations:** MLR assumes linearity, normality, and homoscedasticity, which were violated in this dataset, affecting model accuracy.
- **Multicollinearity:** High correlation among predictors led to unstable coefficients, requiring feature selection and regularization.
- **Limited flexibility:** MLR struggles with nonlinear relationships and complex interactions, reducing its predictive power.
- **Overfitting vs. Underfitting trade-off:** Adding polynomial and interaction terms improved training performance but led to severe overfitting on test data.
- **Computational challenges:** Regularization methods like Lasso, Ridge, and Elastic Net increased complexity and computational cost.
- **Data sensitivity:** MLR is highly influenced by outliers and skewed distributions, requiring robust standard errors and outlier removal.

While MLR provides interpretability, its reliance on strict assumptions and sensitivity to data quality limit its effectiveness compared to more flexible models like Random Forest.

5. Recommendations or conclusions

This analysis provides a foundation for data-driven decision-making in insurance operations by identifying key factors influencing vehicle premium amounts. The business and technical implications, along with conclusions, are discussed in this section.

5.1 Business Implication

As an individual aiming for lower insurance premiums, customers can:

- Choose vehicles with smaller engine capacities to reduce risk-based pricing.
- Set the insured value to reflect actual risk rather than the highest possible valuation.
- Prioritize private-use vehicles over commercial ones, as usage type significantly impacts premiums.

For Insurance Companies:

- Utilize Random Forest models to capture nonlinear risk factors (e.g., engine capacity) and enhance premium pricing accuracy.
    - With insured value contributing over 50% to premiums, accurate vehicle valuation is critical. Regular market updates are essential for competitive and profitable pricing.
    - Engine capacity, the second key factor at about 10%, suggests opportunities for specialized pricing tiers, especially for commercial vehicles.
- Prioritize high-quality input data to improve predictive capabilities.
- Further studies can explore premium optimization or claim estimation models for enhanced decision-making.

5.2 Technical Implication

The study shows Random Forest as a more suitable choice than Multiple Linear Regression (MLR) for capturing nonlinear relationships in insurance pricing, though its effectiveness depends on data quality and computational efficiency.

To enhance accuracy, insurers should consider expanding feature sets by integrating variables such as driver age, telematics data, and regional risk factors to refine risk assessment.

Additionally, improving data processing is crucial for building reliable models. Automating feature engineering can help ensure high-quality data, reducing manual effort and inconsistencies. To address computational challenges, parallel processing and cloud-based computing resources can be leveraged to improve scalability and reduce model training time.

By refining these processes, companies can develop more accurate and scalable pricing models while improving decision-making.

5.3 Conclusion

This study demonstrates that insured value and engine capacity are pivotal in premium pricing, with Random Forest proving to be a robust tool for capturing complex risk relationships. The analysis empowers insurers to adopt data-driven pricing strategies while guiding consumers toward selecting the most cost-effective coverage. Future work should address regional biases and incorporate behavioural data to enhance model generalizability.

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